Yash Bavadiya

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EDUCATION

Indian Institute of Technology, Jodhpur **Aug 2024 – Jul 2028**
Bachelor's Degree - Computer Science and Engineering - **CGPA - 7.21** Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: C++, Rust, TypeScript, JavaScript, Python, C, SQL
Systems & AI: Distributed Systems, Discrete-Event Simulation, Concurrency, OS, Networks, DBMS, RAG, LLM Agents, MCP
Technologies: Qt, React, Node.js, WebAssembly, Cloudflare Workers, PostgreSQL, SQLite, Redis, Docker, Linux, GitHub Actions

EXPERIENCE

Workers IO (YC F26) **Sep 2026 – Present**
Developer Relations Engineer | Deterministic simulation testing Remote (San Francisco)

- Recruited by the founder after building **Breakscale**, an open-source system design simulator on a **discrete-event engine**; now its core maintainer with **2,000+ GitHub stars**, 168 forks, **1,200+ tests** and 31 merged PRs from **15 outside contributors**.
- Shipped **fault injection** (crash, slow, partition, packet loss) and **deterministic replay links**: the same seed and faults replay the exact failure from one shareable URL.
- Built **breakscale-mcp**, an **MCP server** that lets AI assistants model any codebase as a runnable simulation, plus a **VS Code extension**; Breakscale is in the **Vercel**, **Claude** and **Mintlify** open-source programs.

Google Summer of Code 2026 **May 2026 – Aug 2026**
Kdenlive @KDE Community | Open Source Contributor, C++ and Qt Remote

- Designed and shipped **per-channel RGB curves**, a **multi-stop gradient editor** and **time-remap keyframe types** in **C++ and Qt**, all merged for **Kdenlive 26.08**, an editor with **41,000 monthly Flathub installs**.
- Fixed an **11-year-old bug** (reported 2015) with an interval-merging **gap-detection algorithm**; **23 merge requests** merged across Kdenlive and KDE's developer docs after core-maintainer review.

Undergraduate Researcher **Jul 2025 – Nov 2025**
Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder Jodhpur

- Built the retrieval pipeline of an **offline RAG system** over 300-page PDFs with **FAISS** and **4-bit quantized Phi-3**, answering in ∼**3 seconds** on a laptop GPU; benchmarked models and prompts with ROUGE (**+71.77%** over baseline).

PROJECTS

HatchZip | GitHub | Website | **Rust, Slint, ZIP/7z/RAR** **Oct 2026**

- Archive extractor that **frees disk space while it extracts**, by unpacking from the end and truncating the archive; **2,700+ stars**, **27,000+ downloads**, the **#1 most-starred new Rust repo** of its launch week, listed on FMHY next to 7-Zip.
- **Crash-safe by design**: CI kills the process at every write step and verifies every file byte for byte, with **130+ tests** on Linux, macOS and Windows; fixed and published **2 security advisories** from outside researchers.

gyotaku | GitHub | Website | **Rust, ONNX Runtime OCR, SQLite FTS5** **Oct 2026**

- Offline OCR search for your screenshots folder: **1–10 ms per keystroke** across 5,000+ screenshots at 37 MB idle memory; **390+ stars** and **4,000+ downloads** on macOS, Windows and Linux.

toPSD | Website | **Rust to WebAssembly, TypeScript, Cloudflare Workers** **Oct 2026**

- Converts Procreate files to layered Photoshop files **entirely in the browser**; built and shipped in a day, **#1 Product of the Day** on Fazier, with first paying customers from organic search.

Flock | GitHub | npm | **TypeScript, WebSocket, Redis, Docker** **Jun 2026**

- Multiplayer cursors and presence in 10 lines: a horizontally scalable **Redis pub/sub relay** with TTL-based eviction of dead clients; 3 npm packages with **1,100+ downloads** and **58 tests**.

Game Boy Emulator | GitHub | **C++20, SDL2** **Jul 2026**

- Emulator implementing all **501 CPU opcodes**; passes all **11 Blargg CPU test ROMs** and boots Pokémon Red.

ACHIEVEMENTS

- **Open source:** fixes merged into **Deno**, **Adobe React Spectrum**, **Kubernetes SIGs**, **nlohmann/json**, **Vitest**, **Nuxt** and **jj**; maintainer reviewing PRs from 15+ contributors.
- **Building in public: 3,000+ followers** on X, **3.9M impressions** in 3 months; HatchZip's launch post hit **1.9M views**.
- **Coursework:** designed a TAGE-SC branch predictor with **2.9% fewer mispredictions** than ChampSim's best built-in predictor across 10 SPEC CPU 2017 traces.
- JEE Advanced 2024 AIR **3449** of 180,000+; JEE Main **99.65** percentile; **450+** LeetCode problems; Codeforces max **1429**.
- **Junior Core**, DevlUp Labs: mentored 30 students into open source and ran 2 workshops; **Core Member**, Programming Club and E-Cell.